Pakistan Navy Complaint Management System (CMS)

Official Project Documentation & Handover Package

Project Name: Pakistan Navy Complaint Management System (CMS)
Repository: `https://github.com/nexertech/CMS.git`
Current Production Version: `v1.2.4`
Document Generation Date: September 2026

Executive Overview

This documentation package contains the complete formal technical, functional, architectural, and operational documentation suite for the **Pakistan Navy Complaint Management System (CMS)**.

The system is an enterprise-grade, multi-tier web and mobile backend platform built using **Laravel (PHP 8.2)**, **MySQL**, and **RESTful JSON APIs**. It is engineered to streamline, automate, and track the end-to-end maintenance and civic complaint lifecycle across naval housing directorates, Commander Military Engineering Services (CME/CMES), administrative sectors, and field technician workforces.

Documentation Suite Structure

Document Reference	Document Title	Target Audience	Primary Focus
01_URD	**User Requirements Document (URD)**	Executive Leadership, Naval Officers, End Users	Business needs, problem statement, resident and officer workflows, high-level system expectations.
02_SRS	**Software Requirements Specification (SRS)**	Project Managers, System Architects, QA Engineers	IEEE 830 compliant technical specifications, functional & non-functional requirements, data workflows.
03_RBAC	**Role-Based Access Control (RBAC) Manual**	System Administrators, Security Auditors	Security matrix, 19+ granular module permissions, role hierarchy, isolation architecture, and role management.
04_DEV	**Developer & Technical Setup Manual**	Software Engineers, DevOps Engineers	Local environment setup, directory architecture, API contracts, deployment runbook (cPanel/VPS atomic swap).
05_ERD	**Database ERD & Data Dictionary**	Database Administrators, Backend Developers	Visual Entity Relationship Diagrams (Mermaid), 30+ tables schema, foreign keys, indexes, and status mappings.

Document Reference	Document Title	Target Audience	Primary Focus
06_CODE	**Code Commenting & Engineering Standards**	Code Auditors, Development Team	PSR-12 formatting, DocBlocks, architectural commenting guidelines, and critical path audit.
07_OPS	**System User & Administration Manual**	Control Room Operators, Maintenance Staff	Operational walkthroughs: Complaint intake, performa approvals, spare part issuance, resident verification.

Key System Highlights & Architecture Metrics

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|                      PAKISTAN NAVY COMPLAINT MANAGEMENT CMS                   |
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|  Core Framework:     Laravel 12.x on ea-php82 (JazzCloud VPS / Linux cPanel)  |
|  Database Engine:    MySQL 8.0 InnoDB with Granular Foreign Keys & Soft Deletes |
|  Access Control:     19+ Exact-Match Granular Isolated Module Permissions     |
|  Service Workflows:  12 Lifecycle Statuses (TINYINT State Machine)            |
|  Inventory Mgmt:     Spare Parts, Multi-Item Approval Performas, Stock Audit  |
|  Integrations:       Mobile Android/iOS REST APIs, Device Registration, Sanitized|
|  Accessibility:      WCAG AA Compliant Theme Engine (Light, Dark, Night, Auto)  |
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Document Maintenance & Revision Log

Revision	Date	Author / Team	Description
`v1.0.0`	January 2025	Nexertech Core Team	Initial architecture and core complaint ticketing engine.
`v1.2.0`	October 2025	Development Team	State machine refactor: Integer `TINYINT` status migration, approval performa integration.
`v1.2.3`	July 2026	Amjad Subhani (Dev Team)	Mobile API hardening, device management, rate limiting, and session security.
`v1.2.4`	August 2026	Amjad Subhani / Deployment Team	Granular sub-module permission isolation (`Role::hasPermission` exact matching), route middleware sync.
`v1.2.4-DOCS`	September 2026	Engineering Lead	Comprehensive 7-volume documentation and formal handover package compilation.